Function Approximation, Regularization, Ridge, Lasso and Elastic Net

ML & Deep Learning Course

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Outline

1. Function Approximation and Convergence Rates
2. Ordinary Least Squares
3. Ridge Regression
4. Lasso
5. Elastic Net
6. Comparison and Model Selection
7. Summary

The Function Approximation Problem

Setup

Observe n i.i.d. pairs $(X_i, Y_i) \sim \mathcal{D}$, where:

$$Y_i = f^\star(X_i) + \varepsilon_i, \quad \varepsilon_i \overset{iid}{\sim} (0, \sigma^2), \quad X_i \in [0,1]^d$$

Goal: estimate $f^\star$ from data, with no parametric assumption on its form.

Questions:

- How complex is $f^\star$? (smoothness)
- What estimator to use?
- How fast does the error decrease with n?
- What is the best possible rate? (minimax)

Error criterion:

$$\mathcal{R}(\hat{f}) = \mathbb{E}\left[\int_0^1 (\hat{f}(x) - f^\star(x))^2 \, dx\right]$$

(integrated squared risk)

Key insight

The answer depends critically on the **smoothness class** of $f^\star$.

Smoothness Classes

Definition (Hölder Class $\Sigma(\beta, L)$)

For $\beta = k + \alpha$ with $k \in \mathbb{N}_0$ and $\alpha \in (0, 1]$, and $L > 0$, the **Hölder class** is:

$$\Sigma(\beta, L) = \left\{ f : [0,1] \to \mathbb{R} \;\middle|\; \left| f^{(k)}(x) - f^{(k)}(y) \right| \leq L \left| x - y \right|^{\alpha}, \; \forall x, y \right\}$$

Smoothness index β: how many derivatives $f^{\star}$ possesses.

Definition (Sobolev Class $W(\beta, L)$) — Tsybakov, Def. 1.11

For $\beta \in \mathbb{N}$ and $L > 0$, the **Sobolev class** is:

$$W(\beta, L) = \left\{ f \in L^2[0,1] \;\middle|\; f^{(\beta-1)} \text{ abs. cts.}, \; \left\| f^{(\beta)} \right\|_2 \leq L \right\}$$

Ellipsoid characterization: $W(\beta, L) \equiv \Theta(\beta, Q) = \left\{ \theta \in \ell^2 \;\middle|\; \sum_{j=1}^{\infty} j^{2\beta} \theta_j^2 \leq Q \right\}$

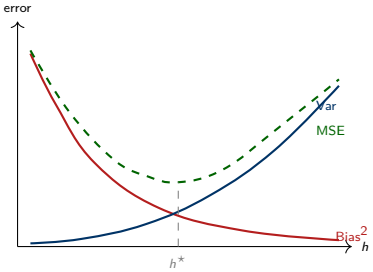
Bias–Variance Decomposition for Estimators

For any estimator $\hat{f}$ of $f^\star$ at a point x_0:

$$\mathbb{E}\big[(\hat{f}(x_0) - f^\star(x_0))^2\big] = \underbrace{\big(\mathbb{E}[\hat{f}(x_0)] - f^\star(x_0)\big)^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{f}(x_0))}_{\text{Variance}}$$

Tuning window h (bandwidth):

- $h \to 0$: use fewer points $\Rightarrow$ high variance, low bias
- $h \to \infty$: use all data $\Rightarrow$ low variance, high bias
- Optimal $h^\star$: balance both



Minimax Convergence Rates

Theorem 1 ((Minimax lower bound — Tsybakov))

For any estimator $\hat{f}$ and $f^\star \in \Sigma(\beta, L)$:

$$\inf_{\hat{f}} \sup_{f^\star \in \Sigma(\beta,L)} \mathbb{E}\left[\left\|\hat{f} - f^\star\right\|_2^2\right] \geq C\, n^{-2\beta/(2\beta+d)}$$

Interpretation

- Curse of Dimensionality and the Smoothness of the Function we are approximating determine the worst possible case.
- Parametric Restrictions alter the sup restriction, allowing to achieve better rates.
- The smoothness hypothesis gives you a class of infinite VC Dimension (Low approximmation error, but high estimation error).

Parametric Models: Recovering Fast Rates

Parametric rate

If $f^\star(x) = \langle \beta^\star, x \rangle$ (linear) with $\beta^\star \in \mathbb{R}^p$ **known**:

$$\left\| \hat{\beta} - \beta^\star \right\|_2 = O_P\left(\sqrt{\frac{p}{n}} \right)$$

$\Rightarrow$ rate $n^{-1/2}$ regardless of d, no curse of dimensionality.

Sparse parametric rate (High-dimensional)

If $\beta^\star$ has at most $s \ll p$ nonzero entries:

$$\left\| \hat{\beta} - \beta^\star \right\|_2 = O_P\left(\sqrt{\frac{s \log p}{n}} \right)$$

Only $s \log p$ "effective parameters" — nearly as good as knowing the support.

This is the key motivation for **Lasso and sparse estimators**: achieve the sparse rate without knowing

Linear Regression: Setup and OLS

Linear model

$$\mathbf{y} = \mathbf{X}\beta^\star + \varepsilon, \quad \mathbf{X} \in \mathbb{R}^{n\times p},\ \mathbf{y} \in \mathbb{R}^n,\ \varepsilon \sim (0, \sigma^2 \mathbf{I}_n)$$

Hypothesis class: $\mathcal{H}_{\text{reg}} = \{x \mapsto \langle \beta, x\rangle : \beta \in \mathbb{R}^p\}$. Loss: $\ell(\beta, (x, y)) = (\langle \beta, x\rangle - y)^2$.

Ordinary Least Squares (OLS)

$$\hat{\beta}_{\text{OLS}} = \underset{\beta\in\mathbb{R}^p}{\arg\min} \|\mathbf{y} - \mathbf{X}\beta\|_2^2$$

Derivation: Set $\nabla_\beta \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = -2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta) = 0$:

$$\mathbf{X}^\top\mathbf{X}\,\hat{\beta}_{\text{OLS}} = \mathbf{X}^\top\mathbf{y} \quad \text{(Normal equations)}$$

If $\mathbf{X}^\top\mathbf{X}$ is invertible: $\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.

OLS: Properties and Failure Modes

Theorem 2 ((Gauss–Markov))

Among all linear unbiased estimators, OLS has the smallest variance (BLUE).

Statistical risk:

$$\mathbb{E}\Big[\Big\|\hat{\beta}_{\mathrm{OLS}} - \beta^\star\Big\|_2^2\Big] = \sigma^2 \operatorname{tr}\big[(\mathbf{X}^\top\mathbf{X})^{-1}\big] = \sigma^2 \sum_{j=1}^{p} \frac{1}{d_j^2}$$

where d_j are the singular values of $\mathbf{X}$.

OLS failure modes

1. $p > n$ **(underdetermined)**: $\mathbf{X}^\top\mathbf{X}$ is singular; infinitely many solutions.
2. **Multicollinearity**: small $d_j \Rightarrow$ huge $1/d_j^2 \Rightarrow$ exploding variance.
3. **No sparsity**: OLS uses all p features; no automatic variable selection.
4. **Overfitting when p is large**: variance term $\sigma^2 p/n \to \infty$.

Ridge Regression: Definition and Derivation

Definition (Ridge / Tikhonov Regularization)

$$\hat{\beta}_{\text{ridge}}(\lambda) = \underset{\beta \in \mathbb{R}^p}{\arg\min} \underbrace{\|\mathbf{y} - \mathbf{X}\beta\|_2^2}_{\text{data fit}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{penalty}}, \quad \lambda > 0$$

Derivation (closed form):

$$\nabla_\beta \left[\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2 \right] = -2\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta) + 2\lambda\beta = 0$$

$$(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\hat{\beta}_{\text{ridge}} = \mathbf{X}^\top\mathbf{y}$$

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{y}$$

- Adding $\lambda\mathbf{I}$ makes the matrix **positive definite** for any $\lambda > 0$, even when $p > n$.
- Constrained form: $\hat{\beta}_{\text{ridge}} = \arg\min_{\|\beta\|_2^2 \le t} \|\mathbf{y} - \mathbf{X}\beta\|_2^2$ (by Lagrangian duality).

Appendix: Singular Value Decomposition (SVD)

The Theorem

Any $n \times p$ centered data matrix $\mathbf{X}$ can be factored as $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, where:

- $\mathbf{U}$ $(n \times p)$: Orthonormal columns ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}$). Left singular vectors.
- $\mathbf{D}$ $(p \times p)$: Diagonal matrix of singular values $d_1 \geq d_2 \geq \ldots \geq d_p \geq 0$.
- $\mathbf{V}$ $(p \times p)$: Orthogonal matrix ($\mathbf{V}^\top\mathbf{V} = \mathbf{I}$). Right singular vectors.

Geometric Interpretation

- **Directions ($\mathbf{v}_j$):** The columns of $\mathbf{V}$ are the principal axes of the feature space.
- **Variance (d_j):** The singular values measure the spread of the data along each axis. A small d_j means the data is nearly flat (low variance) in that direction.
- **Covariance Connection:** $\mathbf{X}^\top\mathbf{X} = \mathbf{V}\mathbf{D}^2\mathbf{V}^\top$. This is the eigendecomposition of the sample covariance matrix.

Ridge: SVD Analysis and Shrinkage

Let $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ (SVD), with singular values $d_1 \geq \ldots \geq d_p \geq 0$.

$$\hat{\boldsymbol{\beta}}_{\text{ridge}} = \mathbf{V}\,\text{diag}\left(\frac{d_j}{d_j^2 + \lambda}\right)\mathbf{U}^\top\mathbf{y} = \sum_{j=1}^{p} \frac{d_j}{d_j^2 + \lambda}\,\mathbf{v}_j\,(\mathbf{u}_j^\top\mathbf{y})$$

Compare with OLS: $\hat{\boldsymbol{\beta}}_{\text{OLS}} = \sum_{j=1}^{p} \frac{1}{d_j}\mathbf{v}_j(\mathbf{u}_j^\top\mathbf{y})$.

Ridge shrinks each component by $d_j^2/(d_j^2 + \lambda)$

- Large d_j: shrinkage factor ≈ 1 (principal directions kept)
- Small d_j: shrinkage factor ≈ 0 (noise directions suppressed)
- All components shrunk, **none set to zero** unless lambda is infinite.

Ridge: Bias–Variance Tradeoff

Bias of Ridge:

$$\mathbb{E}[\hat{\beta}_{\text{ridge}}] - \beta^\star = (\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{X}\,\beta^\star - \beta^\star = -\lambda(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\beta^\star$$

Bias is $O(\lambda)$ — Ridge is **biased** (unless $\lambda = 0$).

Variance of Ridge:

$$\mathrm{Var}(\hat{\beta}_{\text{ridge}}) = \sigma^2(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1} = \sigma^2\sum_{j=1}^{p}\frac{d_j^2}{(d_j^2+\lambda)^2}$$

Variance $\leq$ OLS variance (every term shrunk).

Total MSE

$$\mathrm{MSE}(\hat{\beta}_{\text{ridge}}) = \lambda^2\left\|(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})^{-1}\beta^\star\right\|_2^2 + \sigma^2\sum_{j=1}^{p}\frac{d_j^2}{(d_j^2+\lambda)^2}$$

There always exists $\lambda > 0$ such that $\mathrm{MSE}(\hat{\beta}_{\text{ridge}}) < \mathrm{MSE}(\hat{\beta}_{\text{OLS}})$.

Ridge: Oracle Inequality and Convergence

Theorem 3 ((Oracle Inequality for Ridge — Shalev-Shwartz & Ben-David, Cor. 13.8))

Let the loss be $\ell(w,(x,y)) = (\langle w, x\rangle - y)^2$ *and* $\|x\|_2 \le \rho$, $|y| \le B$. *Then:*

$$\mathbb{E}\big[L_{\mathcal{D}}(\hat{\beta}_{\text{ridge}})\big] \;\le\; \min_{\beta^\star \in \mathbb{R}^p} \Big\{ L_{\mathcal{D}}(\beta^\star) + \lambda \|\beta^\star\|_2^2 \Big\} + \frac{2\rho^2}{\lambda n}$$

Optimal λ and rate

Setting $\lambda = \rho \|\beta^\star\|_2 / \sqrt{n}$:

$$\mathbb{E}[L_{\mathcal{D}}(\hat{\beta}_{\text{ridge}})] - L_{\mathcal{D}}(\beta^\star) \;\le\; \frac{2\rho \|\beta^\star\|_2}{\sqrt{n}} \;=\; O\Big(\frac{1}{\sqrt{n}}\Big)$$

Parametric rate in n, regardless of p — provided $\|\beta^\star\|_2$ is bounded.

Ridge: Geometry of the L2 Constraint

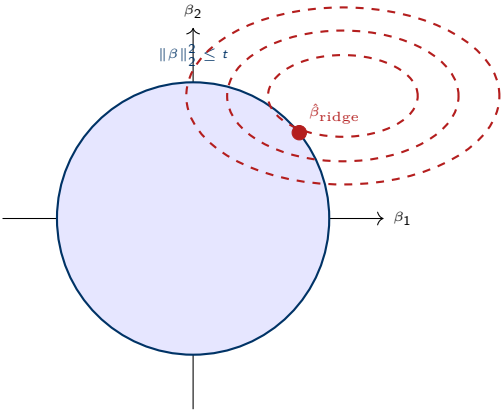
Constrained formulation

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \quad \text{s.t.} \quad \|\beta\|_2^2 \leq t$$

The feasible set $\{\|\beta\|_2^2 \leq t\}$ is a **ball** (smooth, no corners).

Key geometric fact:

- Loss contours (ellipses) expanded until they touch the ball.
- The ball has no corners $\Rightarrow$ the tangent point is **never exactly at an axis**.
- $\Rightarrow$ **Ridge does not produce exact zeros**.



Ridge: Computational Aspects

Direct solution

Solve the linear system $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})\hat{\beta} = \mathbf{X}^\top\mathbf{y}$:

- Cost: $O(p^3)$ via Cholesky decomposition of $(\mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I})$
- When $n \ll p$: use the **dual form** $\hat{\beta} = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top + \lambda\mathbf{I})^{-1}\mathbf{y}$ (cost $O(n^3)$)

SVD-based solution

Compute $\mathbf{X} =^\top$ once $(O(np^2))$, then for any λ:

$$\hat{\beta}(\lambda) = \operatorname{diag}\left(\frac{d_j}{d_j^2+\lambda}\right)^\top \mathbf{y}$$

Entire **regularization path** over all λ at the cost of one SVD.

Ridge: Failure Modes

When Ridge underperforms

1. **No sparsity**: Ridge never sets coefficients to zero.
   When the true $\beta^\star$ is sparse, Lasso may be far more interpretable and efficient.
2. **All variables equally shrunk**: Ridge cannot perform **variable selection**; it keeps all p predictors in the model.
3. **Optimal λ unknown**: must be tuned by cross-validation; wrong λ can lead to severe bias.
4. **Non-linear patterns**: Ridge is still linear; cannot capture interactions without manual feature engineering.

When Ridge excels

- Dense $\beta^\star$ (most features are relevant)
- Highly correlated features (Ridge handles multicollinearity gracefully)
- Prediction only (interpretability not required)

Lasso: Definition and Derivation

Definition (Lasso — Tibshirani, 1996)

$$\hat{\boldsymbol{\beta}}_{\text{lasso}}(\lambda) = \underset{\boldsymbol{\beta}\in\mathbb{R}^p}{\arg\min}\ \frac{1}{2n}\left\|\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right\|_2^2 + \lambda\left\|\boldsymbol{\beta}\right\|_1, \quad \lambda > 0$$

- The ℓ_1 norm $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|$ is **not differentiable** at $\beta_j = 0$.
- Use **subdifferential calculus**: $\partial|\beta_j| = \text{sign}(\beta_j)$ if $\beta_j \neq 0$, $\in [-1, 1]$ if $\beta_j = 0$.

KKT / Subgradient optimality conditions

$\hat{\boldsymbol{\beta}}_{\text{lasso}}$ is a solution if and only if $\forall j$:

$$\frac{1}{n}\mathbf{X}_j^\top(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}) \;=\; \lambda\,\hat{s}_j \quad \text{where} \quad \hat{s}_j \in \begin{cases} \{\text{sign}(\hat{\beta}_j)\} & \hat{\beta}_j \neq 0 \\ [-1,1] & \hat{\beta}_j = 0 \end{cases}$$

Equivalently:

$$\left|\mathbf{X}_j^\top(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}})/n\right| \leq \lambda \quad \text{if } \hat{\beta}_j = 0; \qquad \mathbf{X}_j^\top(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}})/n = \lambda\ \text{sign}(\hat{\beta}_j) \quad \text{if } \hat{\beta}_j \neq 0$$

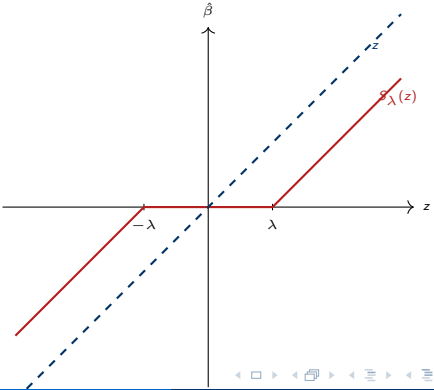
Lasso: Soft Thresholding Operator

For orthogonal design ($\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$), the Lasso has a closed-form solution per coordinate:

$$\hat{\beta}_j = S_{\lambda\lambda}(\hat{\beta}_j^{\mathrm{OLS}}) \quad \text{where} \quad S_{\lambda\lambda}(z) = \mathrm{sign}(z) \cdot (|z| - \lambda)_+ = \begin{cases} z - \lambda & z > \lambda \\ 0 & |z| \leq \lambda \\ z + \lambda & z < -\lambda \end{cases}$$

Interpretation

- Shrinks $\hat{\beta}^{\mathrm{OLS}}$ toward zero by λ
- Sets to **exactly zero** if $|\hat{\beta}^{\mathrm{OLS}}| \leq \lambda$
- Compare: Ridge shrinks by factor $d_j^2/(d_j^2 + \lambda) \neq 0$



Lasso: Geometry of the L1 Constraint

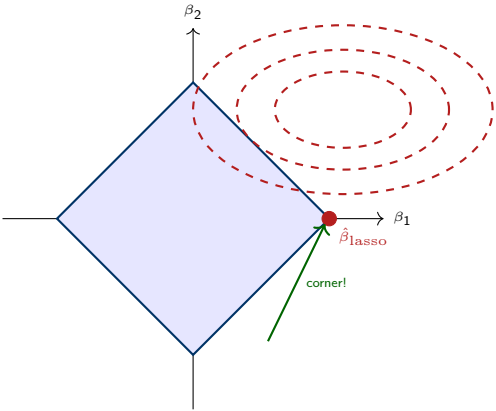
Constrained formulation

$$\min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 \quad \text{s.t.} \quad \|\beta\|_1 \leq t$$

The feasible set is an ℓ_1 **ball** (diamond / cross-polytope).

Why Lasso induces sparsity:

- The ℓ_1 ball has **corners on the axes**.
- Loss ellipse typically first hits the ball **at a corner**.
- At a corner, $\hat{\beta}_j = 0$ for some $j \Rightarrow$ **exact sparsity**.
- In p dimensions: corners at $\pm t\, e_j$, edges, faces.



Lasso: Conditions for Support Recovery

Restricted Eigenvalue (RE) Condition

Let $S = \mathrm{supp}(\beta^\star)$, $|S| = s$. The design $\mathbf{X}$ satisfies $\mathrm{RE}(s, c_0)$ with constant $\kappa > 0$ if:

$$\min_{\substack{\beta \neq 0 \\ \|\beta_{S^c}\|_1 \leq c_0 \|\beta_S\|_1}} \frac{\|\mathbf{X}\beta\|_2}{\sqrt{n}\,\|\beta_S\|_2} \geq \kappa$$

Weaker than RIP; holds for Gaussian designs with high probability when $n \gtrsim s \log p$.

Irrepresentability Condition (Exact Variable Selection)

Let $\underline{C} = \mathbf{X}_{S^c}^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1}$. Exact support recovery requires:

$$\|\underline{C}\ \mathrm{sign}(\beta_S^\star)\|_\infty \leq 1 - \eta \quad \text{for some } \eta > 0$$

Fails when X_{S^c} is **highly correlated** with X_S: irrelevant variables "look like" relevant ones.

Lasso: Convergence Rates

Theorem 4 ((Lasso ℓ_2 convergence rate))

Assume $\mathrm{RE}(s, 3)$ *holds with constant* $\kappa > 0$, $\|\mathbf{X}_j\|_2 \leq \sqrt{n}$ *for all* j. *Set* $\lambda \asymp \sigma\sqrt{\log p/n}$. *Then with probability* $\geq 1 - 2/p$:

$$\left\|\hat{\beta}_{\text{lasso}} - \beta^\star\right\|_2 \leq \frac{4\lambda\sqrt{s}}{\kappa} = O\left(\sigma\sqrt{\frac{s\log p}{n}}\right)$$

Theorem 5 ((Lasso prediction error))

Under the same conditions:

$$\frac{1}{n}\left\|\mathbf{X}(\hat{\beta}_{\text{lasso}} - \beta^\star)\right\|_2^2 \leq \frac{16\lambda^2 s}{\kappa^2} = O\left(\frac{s\log p}{n}\right)$$

Near-oracle rate

The rate $s\log p/n$ is *nearly* the oracle rate s/n (if support were known). The $\log p$ factor is the price of not knowing the support.

Lasso: Failure Modes

1. Highly correlated variables

- Irrepresentability fails: Lasso may select the wrong subset of correlated variables.
- **Instability**: small perturbations in $\mathbf{y}$ can switch which correlated variable is selected.
- Example: if $X_1 \approx X_2$ and both are truly relevant, Lasso arbitrarily selects one.

2. No grouping effect

Lasso does not automatically assign equal coefficients to equal predictors. Elastic Net does.

3. Selection limit: $\min(n, p)$

In the underdetermined case ($p > n$), Lasso can select at most n variables (rank of design).

4. Non-convex variants needed for unbiased selection

Lasso has **shrinkage bias** on the selected coefficients. Post-Lasso OLS or SCAD/MCP penalties can correct this.

Lasso: The Regularization Path

Definition (Regularization path)

The map $\lambda \mapsto \hat{\boldsymbol{\beta}}_{\text{lasso}}(\lambda)$ is a **piecewise-linear** function of λ. As λ decreases from $\lambda_{\text{max}} = \left\|\mathbf{X}^\top \mathbf{y}/(n)\right\|_\infty$:

- $\lambda = \lambda_{\text{max}}$: $\hat{\boldsymbol{\beta}} = \mathbf{0}$ (all zero)
- Coefficients enter the model one at a time at "kink" values of λ
- $\lambda \to 0$: $\hat{\boldsymbol{\beta}} \to \hat{\boldsymbol{\beta}}_{\text{OLS}}$

LARS Algorithm (Efron et al., 2004)

Computes the entire Lasso path in $O(\min(n, p) \cdot np)$ operations — the same cost as a single OLS fit.

Elastic Net: Definition and Motivation

Definition (Elastic Net — Zou & Hastie, 2005)

$$\hat{\beta}_{\text{en}}(\lambda_1, \lambda_2) = \underset{\beta \in \mathbb{R}^p}{\arg\min} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \frac{\lambda_2}{2} \|\beta\|_2^2$$

Equivalently, with mixing parameter $\alpha \in [0, 1]$ and overall penalty λ:

$$\hat{\beta}_{\text{en}} = \underset{\beta}{\arg\min} \frac{1}{2n} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2\right]$$

$\alpha = 1 \Rightarrow$ Lasso; $\alpha = 0 \Rightarrow$ Ridge.

Motivation:

- L1 penalty: encourages **sparsity**
- L2 penalty: handles **correlated variables** (grouping effect)
- Combined: sparse model that respects group structure

Elastic Net: Geometry and Grouping Effect

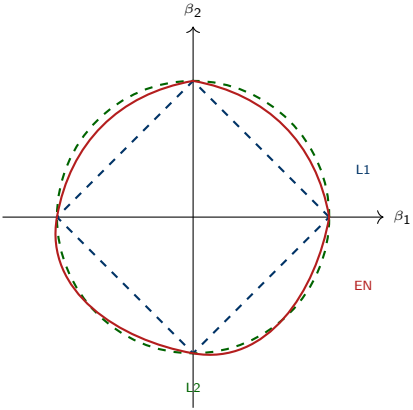
Grouping effect theorem

If $X_i = X_j$ (perfectly correlated), Elastic Net assigns equal coefficients:

$$\hat{\beta}_i = \hat{\beta}_j \quad \text{always}$$

Lasso has no such guarantee — it assigns all weight to one arbitrarily.

The EN penalty ball is a **rounded diamond** — corner singularities are softened by the L2 component, but sparsity is retained.



Elastic Net: Convergence Rate

Theorem 6 ((Elastic Net prediction error))

Under $\mathrm{RE}(s, c_0)$ *on augmented* $\tilde{\mathbf{X}}$*, with* $\lambda_1 \asymp \sigma\sqrt{\log p/n}$ *and* λ_2 *fixed:*

$$\frac{1}{n}\left\|\mathbf{X}(\hat{\beta}_{\mathrm{en}} - \beta^\star)\right\|_2^2 = O\left(\frac{s\log p}{n}\right)$$

Same asymptotic rate as Lasso. L2 penalty improves the ***constant*** *when features are correlated.*

Choosing α

- If features are **uncorrelated**: $\alpha = 1$ (Lasso)
- If features are **grouped and correlated**: $\alpha \in (0.1, 0.9)$ (Elastic Net)
- If all features are **relevant**: $\alpha = 0$ (Ridge)
- **In practice**: tune α by cross-validation from a grid (e.g., $\{0, 0.2, 0.5, 0.8, 1\}$)

Ridge vs. Lasso vs. Elastic Net: Summary

	OLS	**Ridge**	**Lasso**	**Elastic Net**
Penalty	—	$\lambda \|\boldsymbol{\beta}\|_2^2$	$\lambda \|\boldsymbol{\beta}\|_1$	$\lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2$
Solution	Closed form	Closed form	Iterative	Iterative
$p > n$	Fails	Works	Works	Works
Sparsity	No	No	Yes	Yes
Variable selection	No	No	Yes	Yes
Correlated vars	Unstable	Stable	Unstable	Stable
Grouping effect	No	Partial	No	Yes
# hyperparams	0	1 (λ)	1 (λ)	2 (λ, α)
Convergence rate	p/n	$1/\sqrt{n}$	$s \log p/n$	$s \log p/n$

Choosing λ: Cross-Validation

K-fold Cross-Validation for Regularization Path

1. Split data into K folds.
2. For each λ on a grid and each fold k: train on $S \setminus S_k$, evaluate $L_{S_k}(\hat{\beta}(\lambda))$ on fold k.
3. Select $\hat{\lambda} = \arg\min_\lambda \frac{1}{K} \sum_{k=1}^{K} L_{S_k}(\hat{\beta}(\lambda))$.
4. **One-standard-error rule**: select the largest λ within 1 SE of the minimum (more regularization; more parsimonious model).

Information criteria (alternative)

For Lasso with effective degrees of freedom $\hat{df} = |\hat{S}|$ (Tibshirani & Taylor):

$$\mathrm{BIC}(\lambda) = \log\left(\left\|\mathbf{y} - \mathbf{X}\hat{\beta}\right\|_2^2 / n\right) + \frac{\log n}{n} \cdot |\hat{S}|$$

Consistent for support recovery when $\log p/n \to 0$.

High-Dimensional Inference After Selection

Post-selection inference problem

Selecting variables with Lasso and then running OLS on selected variables is **invalid** for hypothesis testing: p-values are anti-conservative due to selection bias.

Approaches to valid post-selection inference

1. **Sample splitting**: use half the data for selection, half for inference.
2. **Debiased / de-sparsified Lasso** (van de Geer et al., 2014):

$$\hat{b} = \hat{\beta}_{\text{lasso}} + \frac{1}{n}\hat{\Theta}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta}_{\text{lasso}})$$

   where $\hat{\Theta} \approx (\mathbf{X}^\top\mathbf{X}/n)^{-1}$. Yields asymptotically normal estimator.
3. **Selective inference / PoSI** (Lee et al., 2016): conditions on the selection event.

Computational Summary and Software

Python: `sklearn`

- `Ridge(alpha=`λ`)`: closed form
- `Lasso(alpha=`λ`)`: coordinate descent
- `ElasticNet(alpha=`λ`, l1_ratio=`α`)`
- `LassoCV`, `RidgeCV`, `ElasticNetCV`: built-in CV
- `lars_path`: LARS regularization path

R: `glmnet`

- `glmnet(X, y, alpha=0)`: Ridge
- `glmnet(X, y, alpha=1)`: Lasso
- `glmnet(X, y, alpha=0.5)`: Elastic Net
- `cv.glmnet`: cross-validated path
- Fortran backend; extremely fast

Warm starting

Computing the path from $\lambda_{\max}$ down to $\lambda_{\min}$ using the previous solution as initialization (**warm start**) reduces total computation to $O(K \cdot n \cdot p)$ for K values of λ.

Summary: Key Take-Aways

Function approximation:

- Smoothness classes control rates
- Minimax rate: $n^{-2\beta/(2\beta+1)}$ in 1D
- Curse of dimensionality: $n^{-2\beta/(2\beta+d)}$
- Sparsity recovers fast parametric rates

Ridge:

- Closed form; handles multicollinearity
- Shrinks all coefficients; **no sparsity**
- Oracle rate $O(1/\sqrt{n})$ under boundedness

Lasso:

- ℓ_1 penalty $\Rightarrow$ exact sparsity (corners)
- Soft thresholding; coordinate descent
- Rate $O(\sqrt{s \log p/n})$ under RE condition
- Fails for highly correlated variables

Elastic Net:

- L1 + L2: sparsity + grouping effect
- Handles correlated groups well
- Equivalent to Lasso on augmented data
- Tune (λ, α) jointly by CV

Next lecture

Random Forests: theory and convergence rates.

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